Constraints Analysis

Job Recruitment and Internship Platform

1. Introduction

This section identifies the technical, business and organizational constraints of the Job Recruitment and Internship Platform, including budget, technology, integrations and deadlines. It is based on the Level 1 context diagram and the MoSCoW-prioritized user stories table.

The platform has three actors (Applicant, Recruiter, Platform Administrator) and two external systems (Email Service and Apache Tika). Main features are CV uploads, job matching, application tracking, interview scheduling and notifications.

Values that are not yet known are marked [confirm] and highlighted in yellow. Replace them with the team's real numbers.

2. Technical Constraints

ID	Constraint	Description / Impact	Source
T1	Apache Tika is the only CV text extractor	CV files must be sent to Tika and the extracted text returned. Job matching depends on Tika being available. Poor extraction from scanned or image-only PDFs lowers matching quality.	Context diagram
T2	Email Service is an external dependency	Sends application confirmations and interview notifications. The platform cannot control delivery; emails may be delayed, fail, or hit rate limits on free plans.	Context diagram
T3	CV file restrictions	Accepted formats (PDF, DOCX), a maximum file size (e.g. 5 MB) and a virus/malware check are needed. Storage must be planned for.	US03, US06
T4	Security and authentication	Hashed passwords, role-based access (Applicant, Recruiter, Admin) and HTTPS. CVs contain personal data, so access must be restricted.	US01, US08
T5	Prototype scope limited to 5 stories	Only US01, US02, US03, US05 and US06 are built. Email, filtering, rejection and admin features are not implemented. The prototype must run end to end without them.	User stories table
T6	Matching must stay simple	Keyword-based matching on Tika text is realistic. AI or ML ranking is out of scope for the prototype because of limited time and data.	Project scope
T7	Performance and availability	Search should respond quickly, and CV extraction should be asynchronous so uploads do not block users. Tika can be slow on large files.	Non-functional
T8	Browser and device support	Must work on current desktop and mobile browsers (responsive web app). Applicants often use phones.	Non-functional

3. Business Constraints

ID	Constraint	Impact
B1	Limited budget	Student project, approx. [confirm]. Only free or open-source tools are used: Apache Tika, free-tier hosting and a free-tier email service. Rules out paid CV-parsing APIs and paid hosting.
B2	Fixed deadline	Deadline: [confirm date]. Forces MoSCoW prioritization: Must Haves first, Could Haves postponed.
B3	Data protection and privacy	CVs and personal data fall under privacy laws (e.g. GDPR or local equivalents). Requires user consent, deletable data, a privacy notice and limited retention.
B4	Fair recruitment	Matching and filtering must not discriminate, so the system should show suitability rather than auto-reject. Affects job matching and US11.
B5	Listing quality	Fake or inappropriate vacancies hurt trust, so moderation by the Administrator is needed. US08 is a Must Have but is excluded from the prototype, which is a business risk to document.
B6	Both sides must benefit	The platform needs enough applicants and recruiters to be useful, and no monetization model is defined for the prototype. Out of scope now but affects long-term viability.

4. Organizational Constraints

ID	Constraint	Impact
O1	Small student team	Team size: [confirm number] members, with limited availability alongside other courses. Limits parallel work and the number of features.
O2	Mixed skill levels	Not everyone may know the chosen stack, Tika or email APIs. Needs task allocation by strength and some learning time.
O3	Coordination	Work must be shared through Git, a task board and regular meetings, with a defined role for each member. Reduces merge conflicts and duplicated work.
O4	Academic requirements	Deliverables must follow course guidelines (diagrams, user stories, documentation, presentation). Documentation takes time away from coding.
O5	No real users or stakeholders	Requirements come from the team and assumptions. Feedback comes late, so requirements may change.
O6	Dependency on third-party services	Team members cannot fix Tika or email provider downtime. Needs fallbacks such as mocks or a local Tika.

5. Integration Constraints

Integration	Direction	Constraint
Apache Tika	Platform sends CV, receives extracted text	Must be deployed (local or Docker). Network latency, file-size limits, text-only output.
Email Service	Platform requests notification, receives confirmation	Needs API key or SMTP setup. Free-tier quotas. A "sent" status is not a guarantee of delivery.

6. Budget and Deadline Overview

Area	Planned approach
Budget	Near-zero: open-source stack, free hosting and email tier.
Phase 1	Must Haves in the prototype (US01, US02, US03, US05, US06).
Phase 2	Should Haves (US07, US11).
Phase 3	Could Haves (US09, US10) and Admin features (US08).
Risk if limits are exceeded	Drop Could Haves first, then Should Haves. Never drop Must Haves in the prototype.

7. Constraint Risks and Mitigation

The table links the main constraints to the risks they create and how the team can reduce them.

ID	Risk	Related constraints	Mitigation
R1	Tika unavailable or poor extraction	T1, O6	Run Tika locally in Docker; fall back to manual keyword entry or a mock extractor.
R2	Email provider limits or downtime	T2, O6	Use a mock email service in the prototype; email is Could Have (US09).
R3	Deadline missed	B2, O1, O4	Follow MoSCoW; drop Could Haves, then Should Haves.
R4	Privacy breach of CV data	T4, B3	Role-based access, HTTPS, consent on upload, deletion option.
R5	Unmoderated fake listings	B5, T5	Admin actions done directly in the database for the prototype; document the risk.

8. Assumptions

- Budget is near zero; only free or open-source tools are used.
- The prototype is a responsive web application implementing US01, US02, US03, US05 and US06.
- Job matching is keyword-based and uses the text returned by Apache Tika.
- Email notifications are shown in the architecture but not built in the prototype.
- Requirements come from the team, as there are no real stakeholders.

9. Issues Found in the Project Documents

While reading the context diagram and user stories, the following inconsistencies were noticed. They should be fixed or justified before submission.

#	Issue	Detail	Suggested action
1	US07 duplicates US05	Both cover editing and deleting job posts.	Merge them or reword one.
2	US04 is missing	Numbering jumps from US03 to US05.	Check whether it was removed and fix the numbering.

#	Issue	Detail	Suggested action
3	US08 (admin) is Must Have but not in the prototype	The teacher may ask why.	Justify in the report, e.g. admin actions are done directly in the database for the prototype.
4	Interview scheduling and job matching have no user story	Both appear in the project idea.	Add stories or state they are out of scope.
5	Email integration vs US09	The diagram shows "Requests email notifications" but US09 is Could Have and not implemented.	Explain that the integration exists in the architecture but is not built in the prototype.

10. Conclusion

The project is mainly limited by a small student team, a fixed deadline, a near-zero budget and its dependence on two external services (Apache Tika and the Email Service). Prioritizing Must Have stories, keeping matching simple, using open-source tools and planning fallbacks for external services keep the prototype realistic and deliverable.